

ANALYSTLAB AFRICA — HEALTHCONNECT EXPERIENCE LAB

Week 7: HealthConnect Analytics Testing & Refinement Report — Data Analytics Track

Prepared by: Omolade

Part 1: Week 6 → Week 7 Transition

Main Week 6 output: A HealthConnect Advanced Analytics & Decision Support Report with statistically validated findings, a composite risk score (0-5), and a CSV handoff file for the Data Science track.

Most important result produced: A composite risk score that separated patients from 11.8% no-show rate (lowest risk) to 81.8% (highest risk).

Main issue/dependency identified: The risk score and handoff file had not yet been tested for reliability across patient subgroups, or for whether the file would actually work cleanly if the Data Science track loaded and used it.

Component requiring testing: The composite risk score (its statistical robustness across segments) and the CSV handoff file (its structural integrity for downstream use).

Relevant track(s): Data Science — direct consumer of the risk score and handoff file.

What was tested: Whether Week 6's KPI calculations were reproducible, whether the risk score held up across age, gender, and appointment-type subgroups, and whether the handoff CSV was genuinely ready for modelling use.

What was expected to improve or be validated: Confirmation that Week 6 figures were correct, and — if any weaknesses were found — an evidence-based refinement, re-tested and validated.

Part 2: Testing Readiness

A. Component Identification

- Week 6 component tested: the composite risk score (0-5) and its handoff CSV file.
- Intended purpose: to give HealthConnect (and the Data Science track) a simple, reliable way to flag high-risk appointments.
- Requirement it should satisfy: the score must be statistically valid, consistent across patient subgroups, and the handoff file must load cleanly with no hidden data-quality issues.
- Tracks connected: Data Science (primary consumer of the handoff file).

B. Testing Objective

The testing this week aimed to answer three questions: (1) Are the Week 6 KPI figures reproducible from the raw data? (2) Does the risk score remain reliable when the data is split into smaller, realistic subgroups? (3) Is the handoff file actually usable by another track without hidden errors?

C. Expected Results (Test Plan)

Test / Scenario	Expected Result	Pass / Fail
Recompute overall no-show rate independently	Should match Week 5/6 reported 48.5%	PASS (48.46%)
Recompute risk score no-show gradient (0-5)	Should match Week 6 reported 11.8% -> 81.8%, increasing	PASS (exact match)
Risk score gradient holds within age_group, gender, appointment_type subgroups	Should increase monotonically in every subgroup	FAIL (broke in 4 of 13 subgroups)
Handoff CSV loads with no missing values in modelling-critical columns	risk_score and other fields should have 0 missing	FAIL (reminder_channel had 1,366 blanks)

Part 3: Testing Results & Refinement

3.1 KPI Recalculation (Pass)

Both the overall no-show rate and the composite risk score were recomputed directly from the raw dataset, independent of the Week 5/6 working files. Both matched exactly (48.46% vs reported 48.5%; identical 11.8%-81.8% gradient). This confirms the underlying calculations were correct.

3.2 Segment Testing — Issue Found

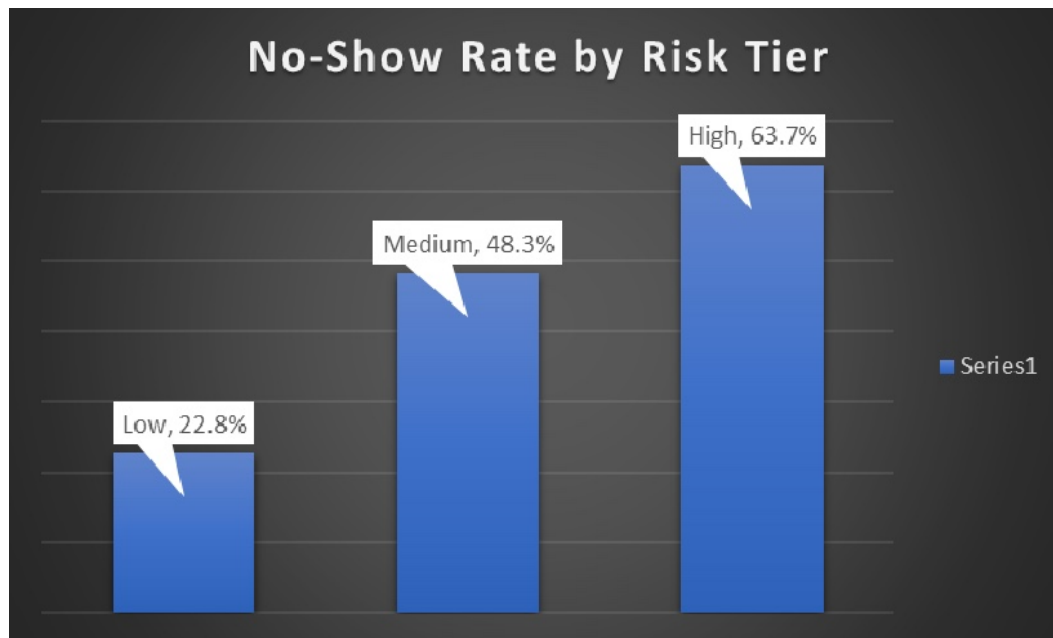
The risk score (0-5) was tested for whether its no-show gradient still increased consistently within smaller subgroups (age group, gender, appointment type). It held up in 9 of 13 subgroups tested, but broke down in 4: the 65+ and 18-24 age groups, the "Prefer not to say" gender category, and Specialist Consultation appointments.

Root cause investigated: In every failing case, the sample size at the extreme risk scores (0 or 5) within that subgroup was very small — as low as 1 to 8 records — making the no-show rate at that single score unstable and easily skewed by a handful of outcomes.

3.3 Refinement: Simplified 3-Tier Risk Band

To fix this, the 0-5 score was collapsed into three broader tiers — Low (0-1), Medium (2-3), High (4-5) — which pools enough records into each group to produce a stable estimate.

Retest result: The simplified 3-tier band was retested across all 13 subgroups (age group, gender, appointment type) and was monotonically increasing (Low < Medium < High no-show rate) in every single one — a full pass, up from 9/13. The overall result across the full dataset is shown below.



Validated improvement: The 3-tier version is the recommended operational form of the risk score going forward — simpler to communicate to clinic staff and more statistically robust than the original 6-point score.

3.4 Cross-Track Testing: Data Science Handoff File (HC-POD Activity)

As this week's mandatory HC-POD cross-track testing activity, the Week 6 handoff CSV was tested as if it were being loaded by the Data Science track for modelling.

Test: Loaded `HealthConnect_RiskScore_Handoff_for_DataScience.csv` and checked for missing values in every column, since unhandled missing data commonly breaks model training.

Finding: `reminder_channel` showed 1,366 missing values. Attempting a first fix — filling the gap with the text "None" — silently failed on retest, because pandas (the tool Data Science would likely use) automatically re-interprets the literal text "None" as a missing value again. This is a genuine, easy-to-miss integration risk.

Action: Refilled the gap using an unambiguous label, "No Reminder Sent", instead of "None".

Retest: Reloaded the corrected file — 0 missing values remained in `reminder_channel`, confirmed via value counts (SMS 2,000 / No Reminder Sent 1,366 / WhatsApp 1,101 / Email 533).

Validated improvement: Published `HealthConnect_RiskScore_Handoff_for_DataScience_v2.csv`, which is now safe to load directly into any standard Python/pandas modelling workflow without hidden null-handling surprises. `distance_to_clinic_km` (90 missing) and `waiting_time_minutes` (60 missing) remain genuinely missing and are flagged for Data Science to handle explicitly (impute or drop) rather than silently mishandled.

Part 4: Updated Business Insights & Recommendations

1. Adopt the simplified 3-tier risk band (Low/Medium/High) operationally instead of the original 0-5 score — it is simpler to communicate and statistically more reliable across every patient subgroup tested.
2. High-tier patients (4-5 points) should remain the top priority for proactive outreach; this holds true across every age group, gender, and appointment type tested.
3. Any file or score shared with another track should be tested as that track would actually use it — the `reminder_channel` issue would not have been caught by only inspecting the file's structure.

4. Continue defaulting to SMS as the primary reminder channel — this Week 5/6 finding was not affected by this week's testing.
5. Before Week 8's final integration, confirm with the Data Science track that v2 of the handoff file resolves their loading concerns, closing the loop on this cross-track fix.

Part 5: Updated Assumptions, Limitations, Risks & Dependencies

- Resolved: the risk score's reliability at fine granularity (0-5) in small subgroups was a previously unknown limitation — now identified, fixed (3-tier band), and validated.
- Resolved: the handoff file's reminder_channel missing-value issue is fixed in v2.
- New limitation discovered: the "Prefer not to say" gender category (n=108) and the 18-24 and 65+ age groups have small extreme-score sample sizes; even the 3-tier band should be interpreted cautiously for these groups until more data accumulates.
- Remaining limitation: distance_to_clinic_km and waiting_time_minutes still contain a small number of genuinely missing values (1.8% and 1.2%); this was flagged for Data Science but not resolved, since it is their decision how to handle it for modelling.
- Dependency: Data Science should confirm receipt and usability of the v2 handoff file before Week 8 final integration.

Part 6: Week 7 Project Summary

Planned to test: The reproducibility of Week 6's KPI calculations, the robustness of the composite risk score across patient subgroups, and the integrity of the Data Science handoff file.

Actually tested: All three — KPI recalculation, risk score segment testing across 13 subgroups, and a full reload test of the handoff CSV.

Most important testing results: KPIs matched exactly on independent recomputation; the 0-5 risk score failed in 4 of 13 subgroups due to small-sample instability; the handoff file had a real, non-obvious missing-value bug.

Issues/weaknesses identified: Risk score instability in small subgroups; a pandas-specific data-loading bug in the handoff file (the string "None" is read back as missing).

Improvements/refinements made: Collapsed the risk score into a 3-tier Low/Medium/High band; relabelled the missing reminder_channel value as "No Reminder Sent" instead of "None".

Retesting results: 3-tier band passed monotonicity testing in all 13 subgroups (up from 9/13); handoff file v2 passed the missing-value check with 0 remaining nulls in reminder_channel.

Track collaborated with: Data Science, via the HC-POD cross-track testing of the handoff file.

What was tested collaboratively: Whether the Week 6 handoff file could actually be loaded and used the way the Data Science track would use it.

What changed as a result: A corrected v2 handoff file was produced and is ready for Data Science to use without encountering the missing-value issue.

Major challenges: The pandas NaN-string interaction was subtle — the first fix attempt looked successful until retested by actually reloading the file, which is why re-testing after every fix mattered.

Important decisions: Chose a 3-tier band over keeping the finer 0-5 score, trading some precision for much greater reliability across all patient groups.

Remaining limitations: Small-sample subgroups (rare gender category, extreme age bands) still deserve caution; distance/waiting-time missing values remain unresolved by design, pending a Data Science decision.

Remaining issues/dependencies: Awaiting confirmation from Data Science that v2 of the handoff file resolves their loading concerns.

Contribution to overall HealthConnect project: Delivered a more reliable, clinic-ready risk indicator and caught a real integration bug before it could affect another track's modelling work.

What must be completed before Week 8: Final confirmation from Data Science on the v2 handoff file, and preparation of Data Analytics' component for final end-to-end integration and presentation.